

Secured Azure VM with RBAC, NSG, Backup & Monitoring

Objective

The objective of this project is to deploy a single-tier Azure Virtual Machine with complete governance, security, backup, and monitoring configurations aligned with AZ-104 core domains. This implementation focuses on establishing enterprise-grade practices for managing identity, networking, backup, and monitoring of Azure resources to ensure compliance, security, and high availability.

Tools & Technologies Used

- Microsoft Azure Portal (Azure pay as-you-go account)
- Windows Server Virtual Machine
- Network Security Group (NSG)
- Azure Recovery Service Vault and Restore Points
- Azure Monitor, Log Analytics Workspace & Alerts
- Azure Role-Based Access Control (RBAC)
- Azure Locks and Tags for Governance

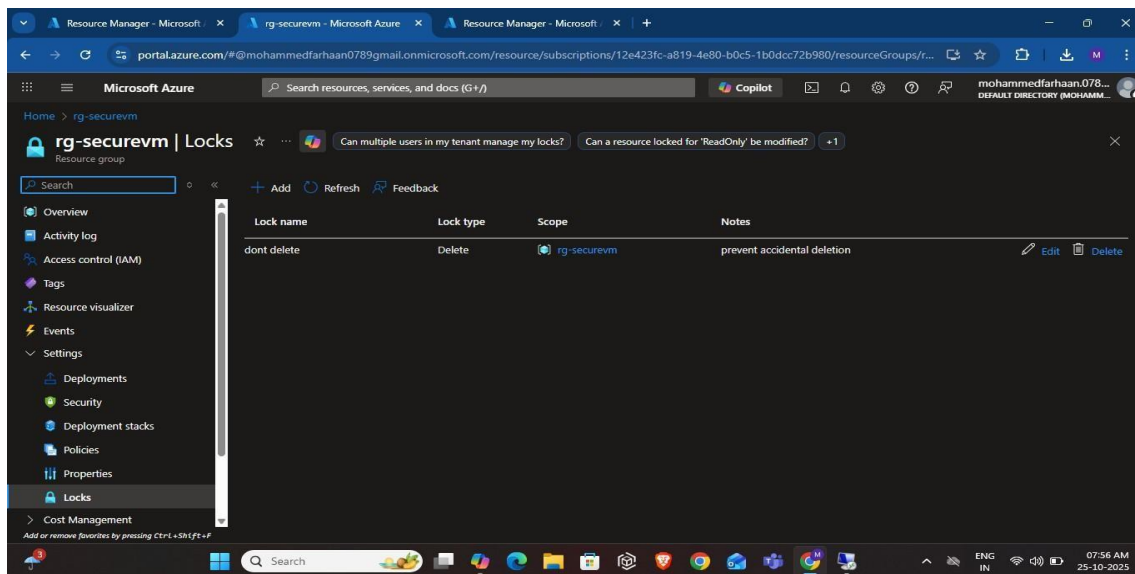
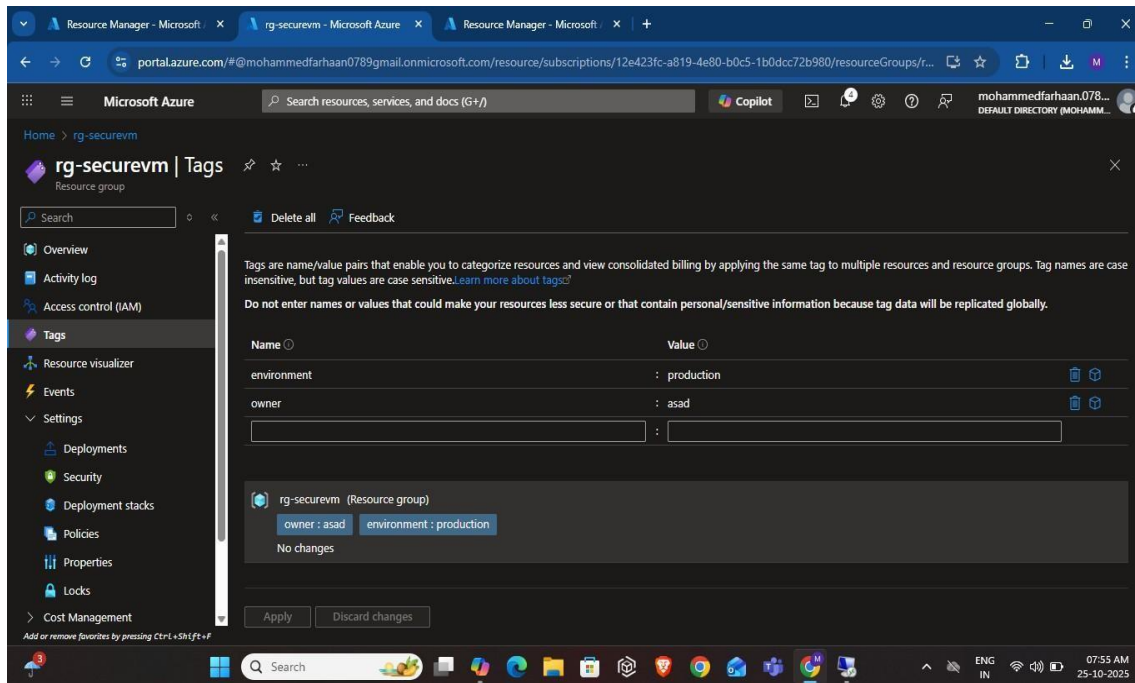
AZ-104 Domains Covered

- Identity & Governance – RBAC, Locks, Tags
- Compute – Virtual Machine, Disks
- Networking – Virtual Network (VNet), NSG, Public IP
- Storage – Recovery Service Vault, Restore Point
- Monitoring – Log Analytics, Alerts

Deployment Steps

1. Resource Group Configuration

A new resource group named 'rg-securevm' was created to logically organize all resources related to this project. Tags were applied to help in cost tracking and categorization. Additionally, a 'CanNotDelete' lock was implemented to prevent accidental deletion of critical resources within the group, ensuring better governance and resource protection.



2.RBAC at the Resource Group Level

A role based access control has assigned at the resource group level named 'rgsecurevm'. A contributor role has assigned to the user01 at a resource group level .A reader role has assigned to the user02 at a resource group level.

Resource Manager - Microsoft | rg-securevm - Microsoft Azure | Resource Manager - Microsoft

portal.azure.com/#@mohammedfarhaan0789gmail.onmicrosoft.com/resource/subscriptions/12e423fc-a819-4e80-b0c5-1b0dcc72b980/resourceGroups/r...

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Home > rg-securevm

rg-securevm | Access control (IAM)

Search

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Locks
Cost Management

Download role assignments | Edit columns | Refresh | Delete | Feedback

Name	Type	Role	Scope	Condition
Owner (2)				
MA Mohammed Asad Farha... 486a854e-5620-4a5a-9...	User	Owner	Subscription (Inherited)	None
MA Mohammed Asad Farha... 486a854e-5620-4a5a-9...	User	Owner	Management group (Inhe...	None
Contributor (1)				
user01 db7888eb-62d4-4ec9-...	User	Contributor	This resource	None
Reader (1)				
user02 d1a24dc7-e880-4d7a-b...	User	Reader	This resource	None
StorageCustomReader (1)				
Unknown fb901c72-d86e-4dbe-...	Unknown	StorageCustomReader	Subscription (Inherited)	None
User Access Administrator (1)				
MA Mohammed Asad Farha... 486a854e-5620-4a5a-9...	User	User Access Administrator	Root (Inherited)	None

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securevm - Microsoft Azure | rg-securevm - Microsoft Azure | Resource Manager - Microsoft

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securevm

Virtual machine

Help me copy this VM in any region | Manage this VM with Azure CLI

Search

Overview
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Application security groups
Network manager
Settings

Help me copy this VM in any region

Connect | Start | Restart | Stop | Hibernate | Capture | Delete | Refresh | Op

Essentials

Resource group (move)	: rg-securevm
Status	: Running
Location	: Central India
Subscription (move)	: Azure subscription 1
Subscription ID	: 12e423fc-a819-4e80-b0c5-1b0dcc72b980

Tags (edit) | Add tags

Properties

Monitoring (8 alerts) | Capabilities (8) | Recommendations | Tutorials

Virtual machine

Networking

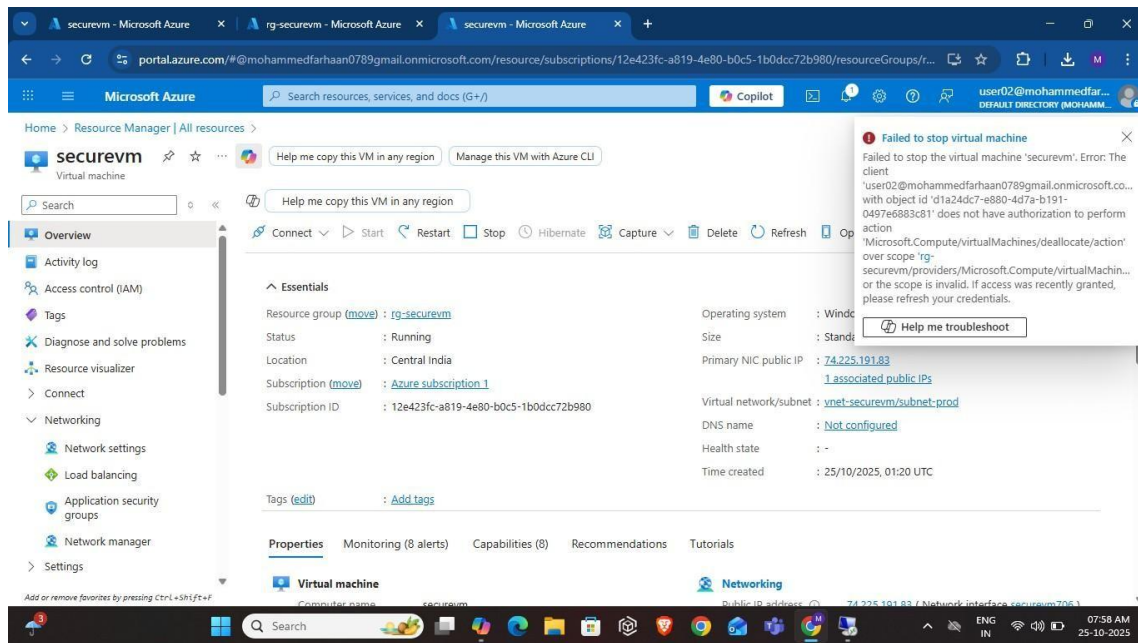
Operating system	: Windows
Size	: Standard B2s (2 vcpus, 4 GiB memory)
Primary NIC public IP	: 74.225.191.83
	: 1 associated public IPs
Virtual network/subnet	: vnet-securevm/subnet-prod
DNS name	: Not configured
Health state	: -
Time created	: 25/10/2025, 01:20 UTC

Failed to delete virtual machine 'securevm'

An error occurred while deleting virtual machine 'securevm' and/or any selected resource(s) associated with it. Error: 'The scope '/subscriptions/12e423fc-a819-4e80-b0c5-1b0dcc72b980/resourceGroups/rg-securevm/providers/Microsoft.Compute/virtualMachines/securevm' cannot perform delete operation because following scope(s) are locked: '/subscriptions/12e423fc-a819-4e80-b0c5-1b0dcc72b980/resourceGroups/rg-securevm'. Please remove the lock and try again.'

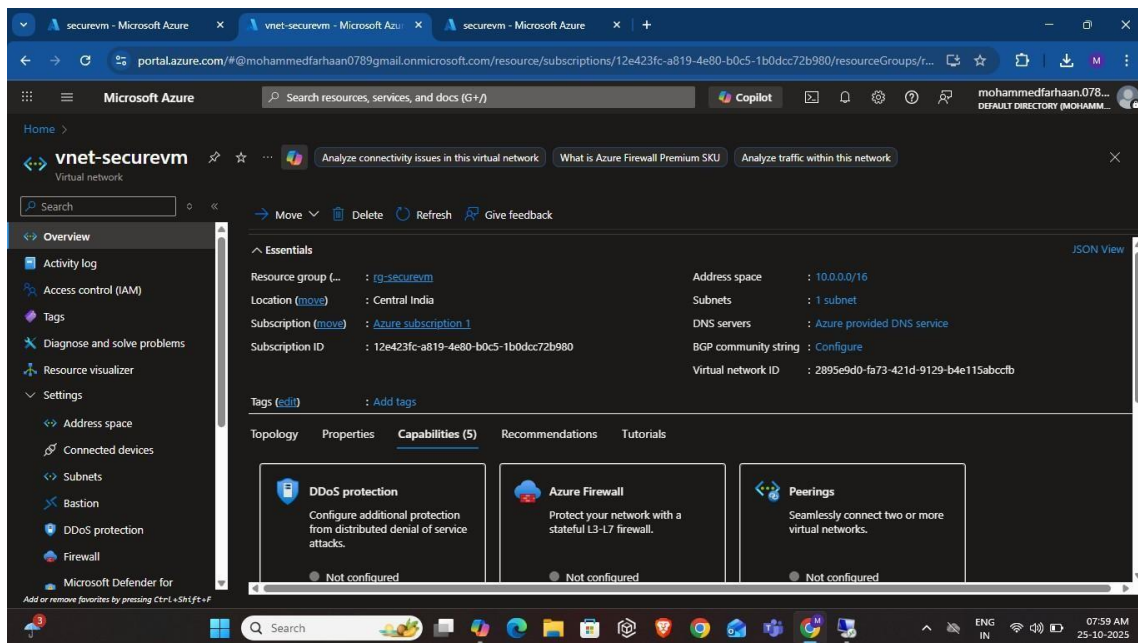
Help me troubleshoot

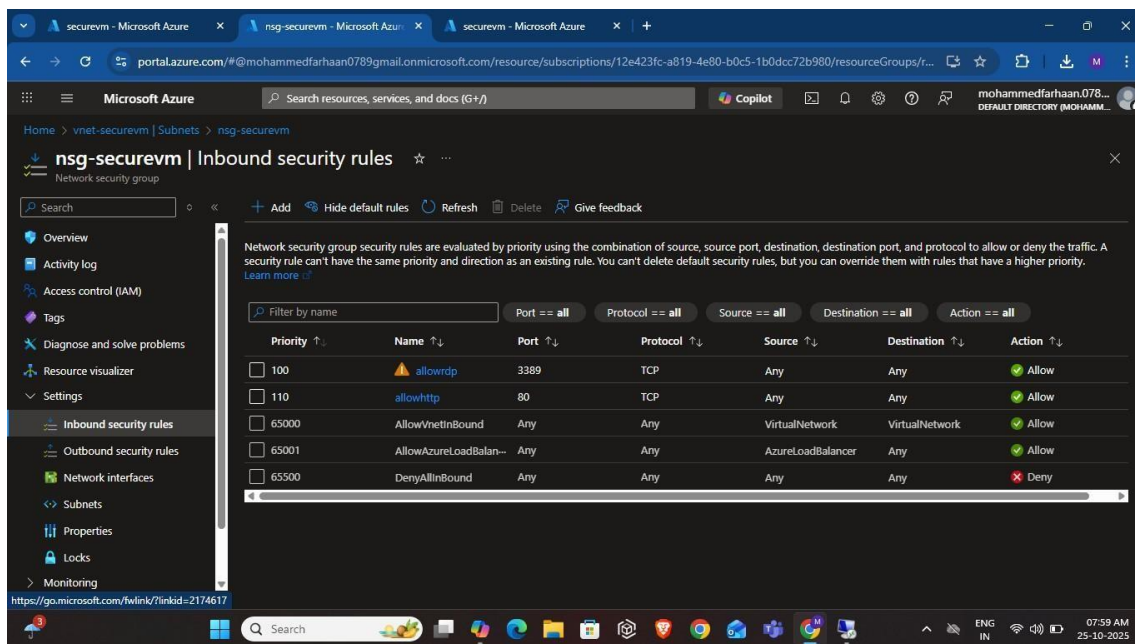
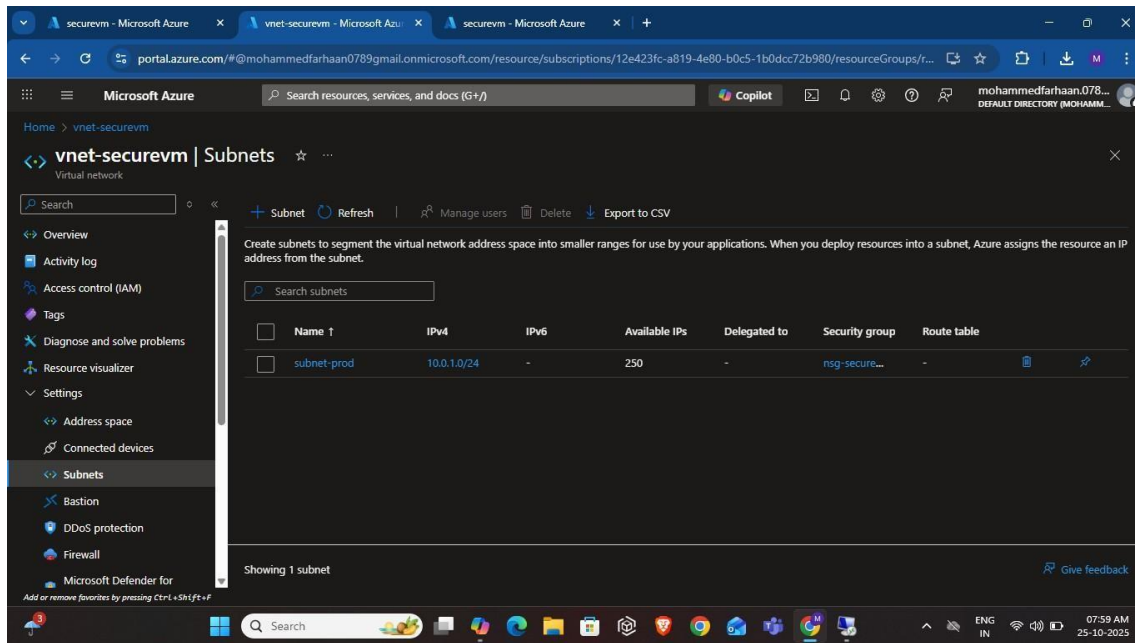
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3. Networking Setup

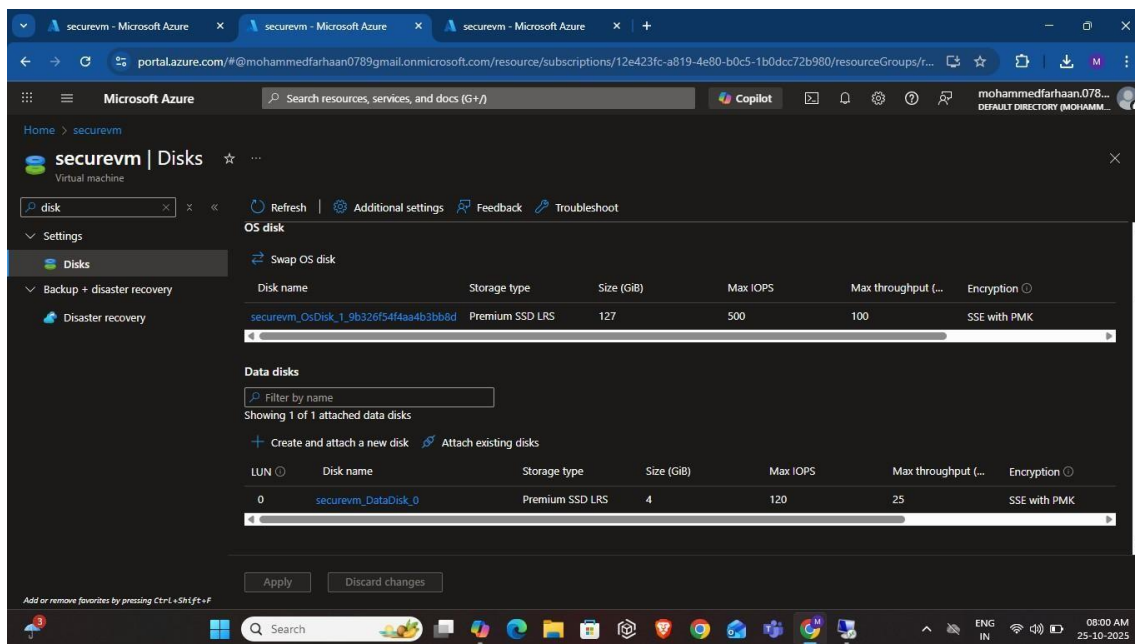
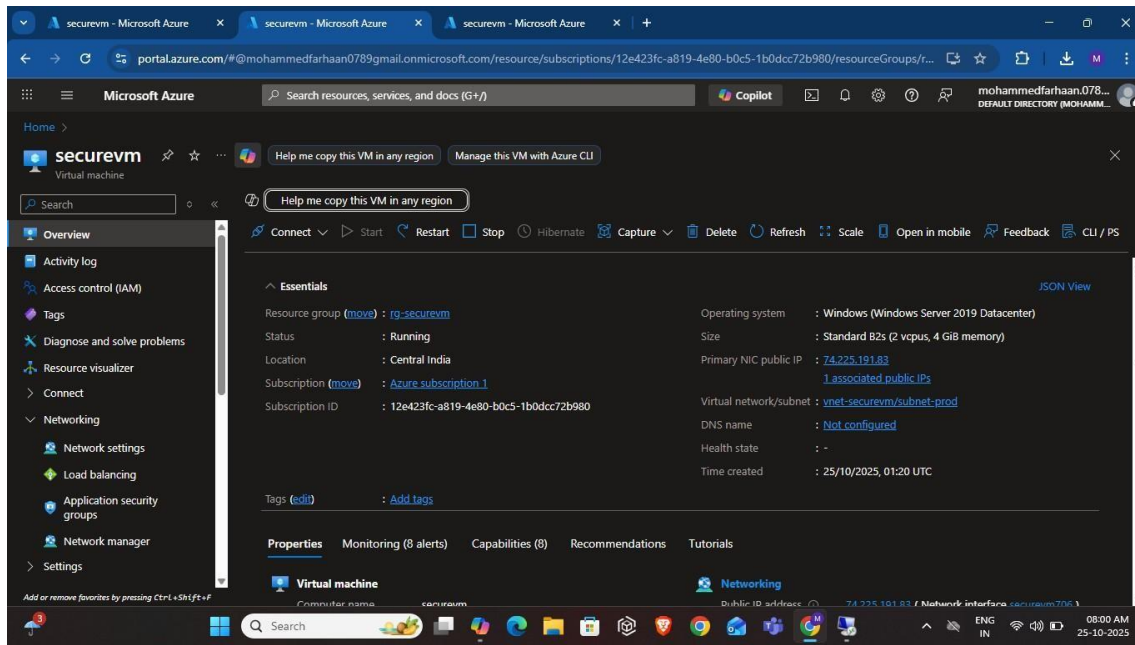
A secure virtual network (VNet) named 'vnet-securevm' was created with subnets for controlled resource communication. An NSG named 'nsg-securevm' was attached to the subnet to define traffic rules. A specific inbound rule for Remote Desktop (RDP) was added to allow administrative access, while other unnecessary ports remained closed. Public IP ('securevm-ip') was assigned to the network interface card (NIC) 'securevm706' to enable external connectivity for management tasks.





4. Virtual Machine Deployment

A Windows Server-based virtual machine named 'securevm' was deployed with both OS and data disks configured. The VM was assigned a system-assigned managed identity to enable secure communication with other Azure services without using credentials. Boot diagnostics and resource monitoring were activated to observe the VM's health and performance.



5. Backup Configuration

A Recovery Services Vault named 'securemvault' was set up to manage the backup and restore configurations. The VM was registered to the vault, and a manual backup job was triggered to create an initial restore point. This ensures that in case of system failure or data corruption, the VM can be restored to a previous healthy state.

securevm - Microsoft Azure x securevm - Microsoft Azure x securevm - Microsoft Azure x +

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Home > securevm

securevm | Backup Virtual machine

back x x < Backup now Restore VM File Recovery Stop backup Resume backup Delete backup data Restore to Secondary Region Undelete ...

Security

- Microsoft Defender for Cloud
- Backup + disaster recovery
 - Backup
- Operations
 - Change tracking

Essentials

Recovery services vault : [securevmvault](#)

Subscription (move) : [Azure subscription 1](#)

Subscription ID : 12e423fc-a819-4e80-b0c5-1b0dcc72b980

Alerts (in last 24 hours) : [View alerts](#)

Jobs (in last 24 hours) : [View jobs](#)

Backup Pre-Check : ✔ Passed

Last backup status : ✔ Success 01/01/2001, 05:30:00

Backup policy : [policyA \(Enhanced\)](#)

Oldest restore point : 25/10/2025, 07:05:48 (56 minute(s) ago)

Included disk(s) : [All disks](#)

JSON View

Recovery points

This list is filtered for last 30 days of recovery points. To recover from recovery point older than 30 days, as well as vault-archive, [click here](#). Long term recovery points can be moved to vault-archive. To move all 'recommended recovery points' to vault-archive tier, [click here](#).

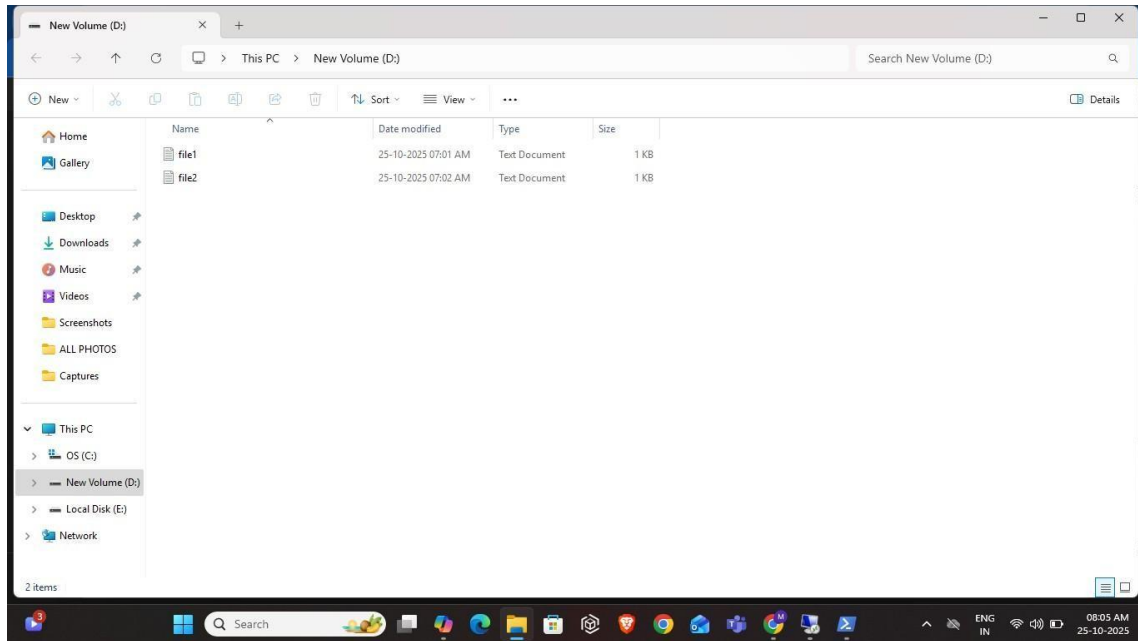
CRASH CONSISTENT	APPLICATION CONSISTENT	FILE-SYSTEM CONSISTENT
0	1	0

Creation time ↑↓	Consistency	Recovery type
25/10/2025, 07:05:48	Application Consistent	Snapshot

https://portal.azure.com/# using Ctrl+Shift+F

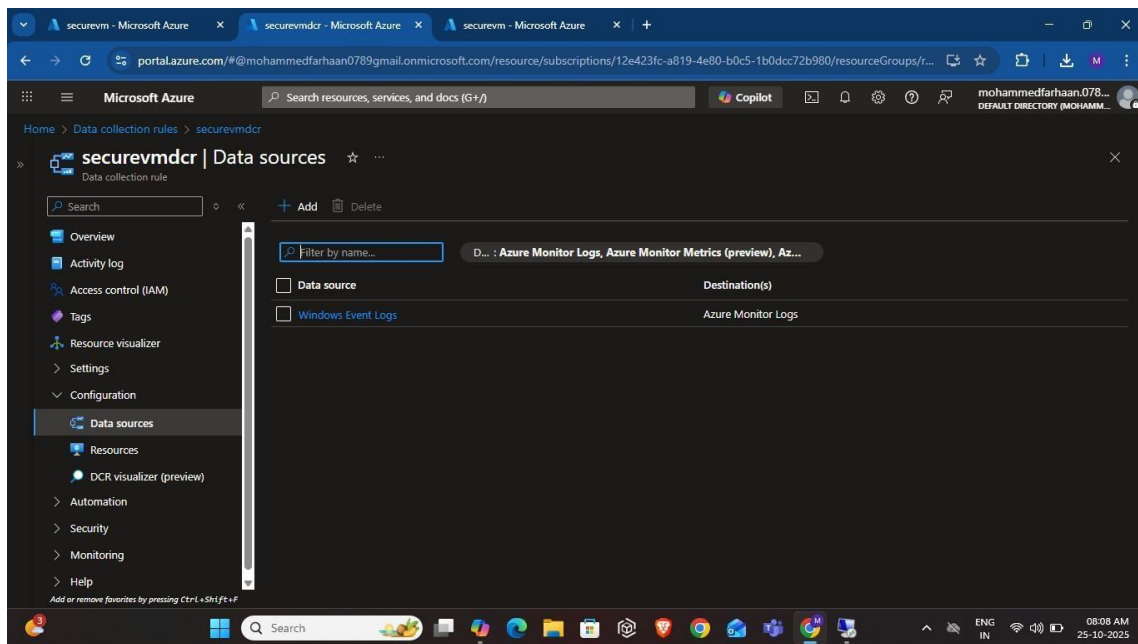
Administrator: Windows PowerShell

```
Checking for OS Compatibility: DONE
Ensuring that the machine meets the requirements to run the recovery script.
Checking for network connectivity: DONE
Checking for required cipher suite: DONE
Checking for Large Disks: DONE
Checking for Storage Pools: DONE
We detected a session already connected to a recovery point of the VM webvmwindows01 .
We need to unmount the volumes before connecting to the new recovery point of VM securevm,
Please enter 'Y' to proceed or 'N' to abort...
Please wait while we disconnect old session...
Older session disconnected. Establishing a new session for the new recovery point...
Connecting to recovery point using iSCSI service...
iSCSI target prepared
Connection succeeded!
Please wait while we attach volumes of the recovery point.
***** Open Explorer to browse for files *****
After recovery, to remove the disks and close the connection to the recovery point, please click 'Unmount Disks' in step 3 of the portal.
```



6. Monitoring and Alerts Setup

Monitoring was enabled by linking the VM to a Log Analytics Workspace named 'securevmlogsanalyticalworkspace'. The Azure Monitor Windows Agent was installed to collect telemetry data. A Data Collection Rule (DCR) named 'securevmdcr' was configured to send performance metrics and logs to the workspace. Alerts were created using Activity Logs and an Action Group to notify administrators about critical changes or performance issues.



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Home > Data collection rules > securevm

securevm | Resources

Overview
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Configuration
Data sources
Resources
DCR visualizer (preview)
Automation
Security
Monitoring
Help

This is a preview version of the new data collection rule associated resources page. Your feedback is appreciated. View the classic experience.

+ Add Disassociate Refresh Edit Data Collection Endpoint

☐ Preselect resources already associated with this data collection rule

Filter for any field... Subscription == Azure subscription 1 Add filter

Resource name	Type	Location	Data Collection...	Resource Group	Subscription
securevm	Virtual machine	Central India	No endpoint co...	rg-securevm	Azure subscript...

Showing 1 - 1 of 1 results.

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securevm - Microsoft Azure | securevmlogsanalyticalworksp: | securevm - Microsoft Azure

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Home > securevmlogsanalyticalworkspace

securevmlogsanalyticalworkspace | Logs

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Agents
Usage and estimated costs
Data export
Network isolation

New Query 1

Run Time range: Last 30 minutes Show: 1000 results KQL mode

1 Event | where EventID == 4798

TimeGenerated [UTC]	Source	EventLog	Computer	EventCategory
> 25/10/2025, 02:23:18.567	Microsoft-Windows-Security-A...	Security	securevm	13824
> 25/10/2025, 02:23:18.564	Microsoft-Windows-Security-A...	Security	securevm	13824
> 25/10/2025, 02:23:16.278	Microsoft-Windows-Security-A...	Security	securevm	13824
> 25/10/2025, 02:23:16.278	Microsoft-Windows-Security-A...	Security	securevm	13824
> 25/10/2025, 02:23:16.278	Microsoft-Windows-Security-A...	Security	securevm	13824

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Microsoft Azure portal interface showing the configuration of an alert rule named "windowseventactivitylogrule".

Alert rules

How many activity log alerts are active: +2

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- History
- Resource visualizer
- Settings
- Alert rule configuration
- Automation
- Help

Scope

Resource: securevm
Hierarchy: Azure subscription 1 > rg-securevm

Condition

Whenever the Activity Log has an event with Category='Administrative'

Actions

Name	Contains actions
windowseventactivitylogalert	1 Email

Showing 1 - 1 of 1 results.

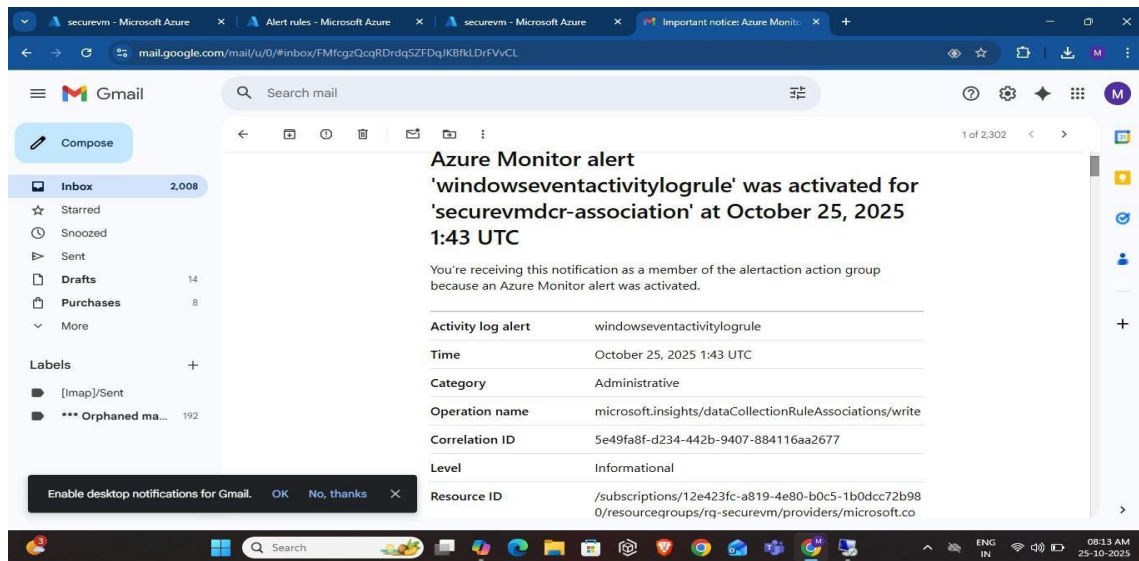
Microsoft Azure portal interface showing the configuration of an action group named "windowseventactivitylogalert".

Action groups

Subscription: Azure subscription 1 | Resource group: all | Location: all | Status: Enabled | Add tag filter | No grouping

Name	Short name	Resource group	Subscription	Actions	Status
windowseventactivitylogalert	alertaction	rg-securevm	Azure subscription 1	1 Email	Enabled

Showing 1 - 1 of 1 results.



7. Validation and Governance Review

After completing all configurations, validation was performed to ensure that NSG rules, RBAC roles, and backup jobs were functioning as expected. The resource locks were verified, and monitoring dashboards confirmed live data collection. This validation ensured that the VM met governance, security, and operational standards as per AZ-104 learning outcomes.

Outcome

The secure Azure VM was successfully deployed with a complete governance and protection model. The project demonstrated practical application of RBAC, NSG hardening, Azure Backup, and performance monitoring. The configuration proved resilient against accidental deletion, ensured system recoverability, and maintained continuous visibility through monitoring tools.

Skills Demonstrated

- Azure Resource Governance using Locks and Tags
- Role-Based Access Control (RBAC) Implementation
- Network Security using NSG Rules
- Azure Recovery Service Vault Configuration and Restore Management
- Performance Monitoring and Log Analytics Setup
- Azure Alerting System using Action Groups

Subscription Used

Azure account (pay as-you-go).